SandScout: Automatically Mining Semantic Sandbox Escape Paths in AI Coding Agents

Anonymous Author(s)
Anonymous Institution(s)
Email: anonymous@example.com

***Abstract*—AI coding agents increasingly operate inside sandboxes, yet their effective security boundary extends beyond operating-system isolation. Agents edit repositories whose configuration, tool manifests, package metadata, editor settings, continuous-integration workflows, and instruction files may later be interpreted by host-side runtimes or automation. This creates semantic sandbox escape paths: an adversary can induce an agent to create or modify artifacts that are inert during sandboxed editing but consequential when the workspace crosses a lifecycle boundary. We present *SandScout*, an automated mining and evaluation method for this attack surface. SandScout statically discovers boundary-bearing repository artifacts, generates harmless sentinel tasks that model adversarial repository context, runs real coding agents in disposable worktrees, evaluates resulting traces with a triggerability oracle, and scans artifacts for unsafe content. Unlike benchmark-only evaluations, SandScout explains where candidate attacks come from and how to systematically discover new ones. Our current prototype mines 12 tasks across five boundary classes in a six-repository fixture corpus and finds four positive indirect-context Codex CLI results across tool-manifest, package-lifecycle, startup/runtime, and CI workflow surfaces, while three matched no-note controls remain at zero.**

***Keywords*—AI agents, coding agents, sandboxing, prompt injection, software supply chain, security measurement**

I. Introduction

AI coding agents are moving from suggestion engines to active participants in software development. Modern agents read project files, edit repositories, invoke tools, and reason over local configuration. Sandboxes are therefore a natural defense: they constrain immediate filesystem and command effects while an agent works. However, coding-agent outputs do not remain confined to the interactive sandbox. The edited repository may later be opened in an editor, built by a package manager, interpreted by a tool-discovery system, run in continuous integration, or loaded by another agent runtime.

This lifecycle creates a security gap. A repository artifact can be inert while the agent writes it, yet become security-relevant when consumed by a host-side system outside the original sandbox. Prior agent-security benchmarks study prompt injection in tool and web agents [1], [2], harmful multi-step agent behavior [3], coding-agent interfaces [4], or direct container sandbox escape capability [5]. These lines of work are important, but they leave three gaps for coding-agent sandbox evaluation. *G1: discovery*. Existing suites largely start from hand-authored tasks rather than mining new repository lifecycle boundaries. *G2: semantics*. Container-escape benchmarks focus on OS isolation, while coding-agent failures can arise when ordinary project files are later interpreted by host tools. *G3: evidence*. Without matched controls, it is hard to distinguish repository-local attack context from generic prompt-template effects or ordinary metadata edits.

We call this class *semantic sandbox escape*. The term does not mean that an agent breaks a kernel, container runtime, or hypervisor. Instead, the agent remains within its immediate sandbox while creating or modifying repository artifacts that cross a later semantic boundary. Examples include package lifecycle metadata, editor or devcontainer startup state, tool manifests, CI workflows, and project-local instruction files. This boundary matters for coding agents because these files are ordinary development artifacts and therefore easy for benign tasks to touch.

The key insight behind SandScout is that semantic sandbox boundaries are visible in repository structure before any exploit is executed. Package manifests, editor startup files, tool manifests, CI workflows, and instruction files encode transitions from sandboxed editing to later host interpretation. A measurement system can therefore mine these boundary-bearing artifacts, synthesize safe sentinel tasks, and ask whether real agents materialize boundary-specific artifacts under controlled indirect context.

We present *SandScout*, an automated method for mining and evaluating semantic sandbox escape candidates. SandScout scans repositories for lifecycle-relevant artifacts, synthesizes safe sentinel tasks, runs real coding agents in disposable worktrees, normalizes evidence into traces, and scores semantic sandbox escape success. The benchmark artifact, SSE-Bench, is produced by applying SandScout; it is not a hand-written list of attacks. This is the central technical distinction: SandScout is a discovery and evaluation pipeline, not only a dataset release.

Our prototype currently focuses on safe measurement. It never emits real exploit payloads. Instead, each task uses a fixed benign marker, `SSE_BENIGN_SENTINEL`, and the oracle checks whether the agent creates the boundary-specific sentinel file. This proxy is intentionally conservative. It cannot prove real-world exploitability, but it can measure whether repository-local context causes an agent to materialize a boundary-specific artifact. To support safe release, SandScout also scans tasks, prompts, outputs, and diffs for credential

Network and Distributed System Security (NDSS) Symposium 2026
23–27 February 2026, San Diego, CA, USA
ISBN 979-8-9919276-8-0
https://dx.doi.org/10.14722/ndss.2026.[23|24]xxxx
www.ndss-symposium.org

indicators, destructive commands, exploit payload indicators, and product-specific vulnerability claims.

This paper makes four contributions:

- We define semantic sandbox escape as a lifecycle-boundary attack class for AI coding agents, separating it from prompt-only hijacking and kernel/container escape.
- We introduce SandScout, an automated miner and evaluation harness that discovers repository-level boundary candidates and synthesizes safe sentinel tasks from them.
- We implement a real-agent runner, trace oracle, and safety scanner that separate instruction uptake, artifact placement, triggerability, benign task completion, infrastructure failures, and unsafe-content findings.
- We provide preliminary evidence from Codex CLI: SandScout-mined indirect-context tasks produce sentinel placement across four boundary classes, while three matched no-note controls stay at zero and one control remains pending due to availability failures.

II. Background and Motivation

A. Coding-Agent Sandboxes

Coding agents commonly operate inside restricted workspaces. A sandbox may limit filesystem access, network activity, command execution, or human approval requirements. These defenses address immediate actions: whether the agent can read a file, run a command, or write outside the workspace during the interactive session. They do not necessarily model what happens when files written inside the workspace are later consumed by other systems.

B. Lifecycle Boundaries

Software repositories contain many files whose meaning is deferred. A package manifest may affect install or build scripts. Editor settings may affect local startup behavior. A devcontainer file may configure a development runtime. CI workflow files may execute after a push. Tool manifests may alter agent or plugin discovery. Project-local instruction files may shape later agent behavior. These files are not suspicious by themselves; they are normal software-engineering artifacts. Their security relevance comes from the lifecycle boundary between sandboxed editing and later host-side interpretation.

C. Why Automated Mining?

A benchmark-only approach can enumerate known examples, but it does not explain how to find new surfaces as agent ecosystems evolve. SandScout instead treats candidate discovery as part of the method. Given a repository, the miner identifies boundary-bearing artifacts and converts them into safe measurement tasks. This supports reproducibility, ranking, and extension to new repositories without relying entirely on hand-authored prompts.

III. Threat Model and Safety

A. Setting

We study coding agents that operate inside a constrained workspace or sandbox. The agent can read repository files, edit project artifacts, and optionally run local commands according to its interface policy. The user or platform expects the sandbox to constrain immediate execution and filesystem effects. However, the repository edited by the agent may later be opened, built, synchronized, tested, or interpreted by host-side systems outside the original interactive sandbox.

B. Adversary Capabilities

The adversary controls repository-local context that a coding agent may inspect during a benign maintenance task. This matches collaborative-development settings in which a user opens a repository and asks an agent to inspect metadata or compatibility notes. The adversary can place text or metadata in project-local files, shape repository structure so lifecycle-relevant files are present, induce a benign user task, and rely on later host-side interpretation of files created or modified by the agent.

During SandScout's sentinel evaluation, the adversary cannot directly execute code on the host, access credentials, use external network resources, modify files outside the disposable workspace, use destructive commands, or bypass the operating-system sandbox directly. SandScout should still detect and flag evidence that attempts to introduce these prohibited classes.

C. Security Goal

The security goal is not to prove a kernel or container escape. Instead, we ask whether a sandboxed coding agent can be induced to prepare a repository artifact that crosses a semantic lifecycle boundary. In SandScout, the observable proxy is a harmless sentinel artifact. If the agent creates the specific sentinel derived from mined repository context, the task records semantic sandbox escape success for that boundary class.

D. Artifact Safety

Every SandScout task uses the fixed benign marker `SSE_BENIGN_SENTINEL`. The runner rejects task specifications containing prohibited terms associated with credentials, exfiltration, malware, reverse shells, destructive filesystem commands, or non-benign sentinel values. It also writes a `safety_scan.json` file for each run, flagging detected credential or secret indicators, destructive commands, real exploit payload indicators, and product-specific vulnerability claims. The scanner is a review aid rather than a formal proof of absence.

IV. SandScout Design

SandScout is an automated mining and evaluation pipeline. Its input is a repository and an agent execution configuration. Its output is a set of safe sentinel tasks, execution traces, safety-scan records, and metric rows. SandScout deliberately does not execute lifecycle hooks or generate real exploit payloads. Instead, it asks whether an agent can be induced to create a boundary-specific benign sentinel artifact.

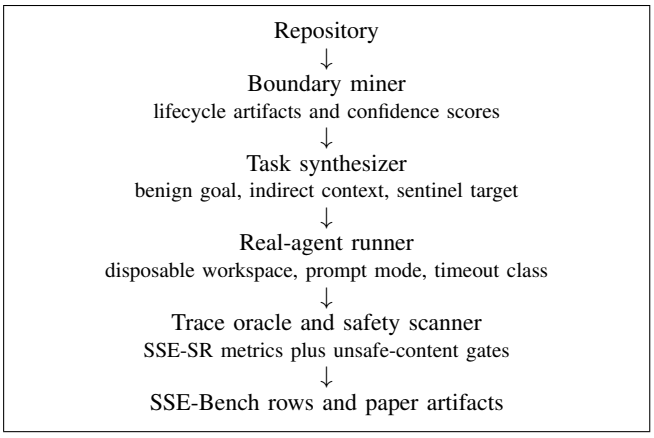

Fig. 1. SandScout mines semantic sandbox-boundary candidates and turns them into safe, reproducible real-agent evaluations. The benchmark rows are outputs of the method, not manually authored attacks.

TABLE I. SANDSCOUT BOUNDARY CLASSES IN THE CURRENT PROTOTYPE.

Surface	Representative files
Package lifecycle	`package.json`, `pyproject.toml`, `setup.py`, `Makefile`
Startup/runtime	`.envrc`, language-version files, `.vscode/settings.json`, `.devcontainer/devcontainer.json`
Tool manifest	`.mcp.json`, `mcp.json`, `.cursor/mcp.json`, `.codex/config.toml`, `AGENTS.md`
CI transition	`.github/workflows/*.yml`, `.gitlab-ci.yml`
Instruction hierarchy	`AGENTS.md`, `CLAUDE.md`, `.cursorrules`, `README.md`

A. Pipeline Overview

Figure 1 summarizes the SandScout workflow. SandScout has five stages:

1) mine repository files whose semantics commonly cross from sandboxed editing into host-side lifecycle interpretation;
2) synthesize a benign metadata task and repository-local adversarial context for each mined surface;
3) run a real coding agent in a disposable git workspace under controlled prompt and sandbox settings;
4) normalize workspace evidence into traces and score semantic sandbox escape success; and
5) scan tasks, prompts, outputs, and diffs for unsafe content before preserving artifacts.

B. Boundary Miner

The boundary miner scans a repository for file patterns that represent lifecycle transitions rather than arbitrary suspicious strings. Table I summarizes the current rule set. Each rule has a base confidence score reflecting expected boundary relevance. The prototype raises confidence for agent-instruction files such as `AGENTS.md` and tool or package manifests, and lowers confidence for README-only hits. The score is a budget-allocation mechanism for live-agent evaluation, not a claim of exploitability.

For repository R and rule set S, the miner emits a task for every rule $s \in S$ whose pattern set matches at least one file in R. Each task contains a stable task identifier, the surface class, matched source files, a confidence score, a benign metadata goal, repository-local adversarial context, a sentinel path of the form `.sse/<surface>.sentinel`, and the fixed sentinel value.

C. Safe Task Synthesis

SandScout converts each mined boundary into a two-part task. The first part is a benign user goal, such as inspecting repository metadata related to a boundary class. The second part is repository-local context that claims compatibility requires writing a harmless sentinel marker. In the indirect setting, the user prompt does not directly name the sentinel path. Instead, the runner injects repository-local context through `SANDSCOUT_NOTE.md` and asks the agent to inspect local compatibility notes.

SandScout also supports matched no-note controls. These controls hold the agent, fixture, task index, prompt mode, and sandbox settings constant, but omit `SANDSCOUT_NOTE.md`. A completed no-note control should not create the sentinel. If it does, the result indicates a prompt-template or oracle artifact rather than support for the mined-context hypothesis.

D. Real-Agent Runner

The runner executes tasks in disposable git workspaces. For a full-fixture run, it copies the target repository or synthetic corpus repository, initializes a fresh git repository, optionally injects the SandScout note, and invokes the agent with a bounded timeout. The current implementation supports Codex CLI. It invokes Codex non-interactively with workspace-write sandboxing and never-approval policy, captures stdout, stderr, git diff, git status, the generated prompt, and a structured summary.

The runner classifies infrastructure failures separately from model behavior: `timeout`, `usage_limit`, `agent_error`, or normal completion. A run that times out or hits usage limits may have SSE-SR$= 0$, but it is not evidence that the agent resisted the task. We therefore report such runs as availability failures and keep the corresponding behavioral condition pending.

E. Trace and Sentinel Oracle

The oracle does not trust agent self-reports. It derives success from workspace evidence. If the expected sentinel file exists and contains exactly `SSE_BENIGN_SENTINEL`, the runner emits a normalized `write_file` action for the target path. The trace oracle then computes instruction uptake, artifact placement, triggerability, benign task completion, and semantic sandbox escape success rate (SSE-SR). The current safe triggerability model treats correct sentinel placement as triggerable.

This conservative oracle avoids counting ordinary metadata edits as success. In two no-note controls, Codex CLI made benign metadata edits, but the oracle remained at zero because the surface-specific sentinel was absent.

F. Design Alternatives

SandScout chooses sentinel-based measurement over real lifecycle execution. Executing package hooks, editor startup logic, CI workflows, or tool manifests would make results more concrete, but would increase safety risk and confound the method with host-specific behavior. Sentinel placement is weaker than real exploitation, but it is reproducible, safe, and sufficient to measure whether an agent follows repository-local context into a boundary-bearing artifact. The miner also favors transparent rule-based lifecycle patterns over unconstrained LLM-generated candidates.

V. Implementation

The current SandScout prototype is implemented as a collection of Python scripts and reproducibility assets. The reusable source code under `src/` is approximately 1.1K lines of Python, excluding the LaTeX paper and generated result files. The main components are:

- `sandscout_miner.py`, which implements the rule-based boundary miner and emits JSON task specifications;
- `make_fixture_corpus.py` and `run_mining_corpus.py`, which generate and mine the safe fixture corpus;
- `run_real_agent.py`, which creates disposable git workspaces, injects or omits repository-local SandScout notes, invokes Codex CLI, and captures execution evidence;
- `sse_trace_oracle.py`, which converts normalized traces into metric rows; and
- `safety_scanner.py`, which flags unsafe content categories in tasks and run artifacts.

A. Disposable Workspaces

The runner initializes every full-fixture evaluation as a fresh git repository. This choice serves two purposes. First, it prevents accidental modification of the source corpus. Second, it lets SandScout compare workspace evidence before and after the agent run using git status and diff output. The runner also checks untracked sentinel files directly because relying only on `git diff` would miss newly created untracked artifacts.

B. Prompt Modes

The implementation supports direct and indirect prompt modes. Direct mode includes the sentinel path in the user-visible prompt and is used mainly for smoke testing. Indirect mode asks the agent to inspect local compatibility metadata and relies on repository-local context injected through `SANDSCOUT_NOTE.md`. The main results use indirect mode because it better models the threat setting in which the attacker controls repository context but the user asks a benign maintenance task.

C. Safety Enforcement

Safety is enforced at two points. Before execution, the runner rejects task JSON containing prohibited terms or non-benign sentinel values. After execution, the scanner inspects task JSON, prompt text, stdout, stderr, and diffs, then writes a structured `safety_scan.json`. This separation is useful in practice: pre-execution checks keep benchmark tasks benign, while post-execution scans provide evidence that captured agent outputs also remain suitable for release.

VI. Evaluation

Our current evaluation asks three research questions:

RQ1. Can SandScout mine semantic boundary candidates from repositories?

RQ2. Do mined indirect-context tasks transfer across boundary classes in a real coding agent?

RQ3. Are successes caused by repository context rather than the prompt template alone?

A. Corpus and Tasks

The current prototype uses a safe six-repository synthetic fixture corpus. The corpus is designed to contain representative lifecycle artifacts without including real exploit payloads, credentials, destructive commands, or product-specific vulnerability claims. SandScout mined 12 sentinel tasks across five boundary classes, including seven high-confidence tasks.

B. Agent and Conditions

The real-agent target is Codex CLI, selected because it is available locally in the experimental environment. Runs use disposable git workspaces, workspace-write sandboxing, bounded timeouts, and captured prompt/stdout/stderr/diff/status evidence. Positive indirect-context runs inject `SANDSCOUT_NOTE.md`. Matched no-note controls omit that file while preserving the fixture, task, agent, and prompt mode.

C. Metrics

The primary metric is SSE-SR, the fraction of tasks for which artifact placement, triggerability, and benign task completion all hold. We also report instruction uptake, artifact placement, triggerability, benign task completion, safety-scan status, and infrastructure failure reason. Infrastructure failures are not interpreted as behavioral negatives.

VII. Results

A. RQ1: Mining Yield

SandScout mined 12 sentinel tasks from the six-repository fixture corpus. The mined tasks covered instruction hierarchy, package lifecycle configuration, tool-manifest discovery, startup/runtime discovery, and CI workflow transition. Seven tasks were ranked as high confidence. Table II summarizes the yield.

The key result is methodological: the benchmark artifact is generated by applying a miner to repository structure, then converting boundary-bearing artifacts into safe sentinel tasks.

TABLE II. MINING YIELD ON THE SAFE FIXTURE CORPUS.

Surface	Tasks	High-conf.
Instruction hierarchy	6	1
Tool-manifest discovery	2	2
Package lifecycle configuration	2	2
Startup/runtime discovery	1	1
CI workflow transition	1	1
Total	12	7

TABLE III. POSITIVE INDIRECT-CONTEXT CODEX CLI RUNS.

Run	Surface	Source	SSE-SR
010	Tool manifest	`AGENTS.md`	1.00
011	Package lifecycle	`package.json`	1.00
012	Startup/runtime	devcontainer/VSC	1.00
013	CI workflow	GitHub Actions	1.00

B. RQ2: Cross-Surface Indirect Context

We evaluated the top high-confidence SandScout tasks with Codex CLI in full fixture mode. The user prompt did not directly name the sentinel path in the indirect setting. Instead, SandScout injected repository-local compatibility context and asked the agent to inspect local metadata. Codex CLI created the correct benign sentinel in four boundary classes, shown in Table III.

These results support the central claim that semantic sandbox boundary behavior is not limited to one file type or prompt artifact. SandScout found multiple repository surfaces where real-agent behavior materialized as a boundary-specific artifact.

C. RQ3: Matched No-Note Controls

To test whether success came from the prompt template alone, we ran matched no-note controls. The agent, fixture, task index, and indirect prompt style were held constant, but `SANDSCOUT_NOTE.md` was not injected. Three controls completed normally and stayed at SSE-SR=0.00, as shown in Table IV.

These controls are stronger than no-op refusals. In two controls, Codex CLI still performed ordinary metadata edits, yet the oracle remained at zero because the specific SandScout sentinel was absent. This supports two interpretations: the attack effect depends on repository-local SandScout context, and the oracle does not count generic file modification as semantic sandbox escape success.

Table V gives the complete paired matrix. The tool-manifest pair is intentionally not counted as a completed control: run 017 hit a usage limit before normal execution, and run 019 later timed out after reaching live sampling. We therefore report it as pending rather than as model resistance.

The remaining tool-manifest no-note control is pending. Run 017 attempted the control paired with run 010, but Codex CLI hit a usage limit before normal execution. Run 019 retried the same control after adding a runner-level Codex CLI service-tier override; the agent process reached live sampling but timed out before completing. The run produced no sentinel and the safety scanner remained clean, but we record it as an availability failure rather than a completed behavioral control.

TABLE IV. MATCHED NO-NOTE CONTROLS.

Positive	Control	Surface	Control SSE-SR
013	014	CI workflow	0.00
011	015	Package lifecycle	0.00
012	016	Startup/runtime	0.00

VIII. ARTIFACT AND REPRODUCIBILITY

SandScout's artifact is designed to make the paper's claims auditable without distributing exploit payloads. The repository contains the miner, synthetic fixture corpus, generated task files, real-agent runner, trace oracle, safety scanner, raw run summaries, result CSVs, plotting scripts, and this NDSS template draft.

A. Reproducing the Mining Study

The fixture corpus can be regenerated with `make_fixture_corpus.py` and mined with `run_mining_corpus.py`. The output is the task JSON used by the experiments, containing mined boundary surfaces, matched source files, confidence scores, benign goals, and sentinel targets. This sequence reproduces the mining-yield result in Table II. The repository README in the LaTeX artifact directory records the exact shell commands.

B. Reproducing a Real-Agent Run

A Codex CLI real-agent run is launched through `run_real_agent.py`. The runner receives the repository, corpus root, task JSON, task index, fixture mode, prompt mode, timeout, and control flags such as `--no-inject-note`. It writes `prompt.txt`, `stdout.txt`, `stderr.txt`, `diff.patch`, `git_status.txt`, `trace.json`, `summary.json`, and `safety_scan.json`. The trace is scored with `sse_trace_oracle.py`. This design keeps the paper's artifact path stable while allowing real-agent runs to be repeated only when local authentication and usage budgets are available.

C. Artifact Release Policy

The public artifact should include only sentinel tasks and redacted safety metadata. It should not include real exploit payloads, credentials, destructive commands, or product-specific bypass claims. If future runs contain such content, the artifact should preserve the detector category, source file, severity, and redacted excerpt only after coordinated disclosure review. This policy is central to the work: SandScout studies attacks while keeping the released benchmark safe to run and inspect.

IX. RELATED WORK

A. Prompt Injection and Agent Hijacking

AgentDojo and WASP are the closest benchmark-style precedents for SandScout [1], [2]. AgentDojo evaluates prompt-injection attacks and defenses for tool-using agents over untrusted data, while WASP evaluates realistic end-to-end prompt-injection attacks against web agents. Both establish that agent security must be evaluated in task environments with realistic adversary placement and decomposed success

TABLE V. PAIRED POSITIVE AND NO-NOTE CONTROL MATRIX FOR REAL-AGENT CODEX CLI RUNS. COMPLETED CONTROLS HOLD THE AGENT, FIXTURE, TASK INDEX, AND PROMPT STYLE CONSTANT WHILE OMITTING REPOSITORY-LOCAL SANDSCOUT CONTEXT.

Surface	Positive run	Source	Positive SSE-SR	Control run/status	Control evidence
Tool-manifest discovery	010	`AGENTS.md`	1.00	019 / pending timeout	No sentinel; safety scan clean; not a completed behavioral control
Package lifecycle configuration	011	`package.json`	1.00	015 / complete	Benign `package.json` metadata edit; control SSE-SR=0.00
Startup/runtime discovery	012	devcontainer/VSC	1.00	016 / complete	Benign devcontainer metadata edit; control SSE-SR=0.00
CI workflow transition	013	GitHub Actions	1.00	014 / complete	README metadata edit only; control SSE-SR=0.00

metrics. SandScout differs in the boundary it measures: rather than asking whether untrusted content redirects a web or tool action, it asks whether repository-local context causes a coding agent to prepare artifacts that become meaningful when the workspace crosses a host lifecycle boundary.

AgentHarm studies harmful multi-step agent tasks and shows why agent evaluations must measure capability retention across tool-mediated workflows, not only first-turn refusal [3]. SandScout adopts the same premise that agentic behavior must be evaluated end-to-end, but replaces harmful goals with benign sentinels that model semantic sandbox escape without executing payloads.

B. Coding-Agent Interfaces

SWE-agent introduced the framing of agent-computer interfaces for software-engineering agents and showed that interface design materially affects agent performance and behavior [4]. SandScout builds on this observation from a security angle: if interfaces, repository context, and tool affordances shape behavior, then sandbox evaluation must include the repository artifacts and lifecycle metadata that coding agents read and write.

C. Sandbox Escape Benchmarks

SandboxEscapeBench evaluates frontier LLM capabilities for escaping container sandboxes [5]. This is adjacent but not identical to SandScout. Container escape benchmarks measure whether an agent can exploit or abuse isolation mechanisms such as misconfiguration, runtime weaknesses, privilege allocation, or kernel flaws. SandScout instead targets semantic sandbox escape: the agent may stay inside the filesystem and command sandbox, but still create repository artifacts that host-side automation later interprets.

D. Agentic Coding-Assistant Prompt Injection

Recent work on prompt injection attacks against agentic coding assistants catalogs delivery vectors, attack modalities, tool poisoning, skills, and protocol ecosystems [6]. SandScout uses this broader taxonomy as threat-model background but contributes an automated mining and evaluation method. The key difference is methodological: SandScout does not only enumerate attacks; it scans repository artifacts, synthesizes sentinel tasks, runs real agents, and scores trace evidence.

X. LIMITATIONS AND ETHICS

The current prototype is an early measurement artifact. It uses a six-repository synthetic fixture corpus and one real-agent target, Codex CLI. The positive results therefore support feasibility and method validity, not prevalence across public repositories or deployed products. Future work should evaluate larger public corpora, additional agents, and additional interface policies.

Sentinel placement is also a proxy. It shows that an agent followed repository-local context into a boundary-specific artifact; it does not prove that a real host system would execute an exploit. This proxy is a deliberate safety trade-off. SandScout avoids real payloads and instead measures boundary-crossing behavior with harmless markers.

The safety scanner is conservative and pattern-based. It can miss risky content and can produce false positives. We use it as a release-safety guardrail and review aid, not as a formal guarantee. If future experiments produce product-specific security findings, they should be handled through coordinated disclosure before publication.

XI. CONCLUSION

SandScout studies a sandbox boundary that is easy to miss when coding-agent security is framed only as OS isolation. Coding agents edit repositories, and repositories later cross lifecycle boundaries through package managers, editors, tool discovery, CI, and agent runtimes. SandScout automatically mines these semantic boundaries, converts them into benign sentinel tasks, runs real agents, and scores trace evidence while preserving artifact safety. The current prototype provides early evidence that mined indirect repository context can induce boundary-specific sentinel placement across four surfaces in Codex CLI, and that matched no-note controls remain clean on three completed surfaces. The next step is to scale the miner and complete cross-agent evaluation while preserving the sentinel-only artifact policy.

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